SmartDesk V2: Enterprise-Grade IoT Workstation Controller

Dual-Tier Connectivity, Autonomous Captive Portal Bypass, and Edge RTOS Design

SmartDesk Architecture Team

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Key Highlights: Autonomous Fortinet Gateway Bypass • Offline BLE 5.0 • Render WSS Cloud • FreeRTOS Hardware Spinlocks • Edge Schedules

Project Motivation & Industry Bottlenecks

Conventional IoT Pitfalls:

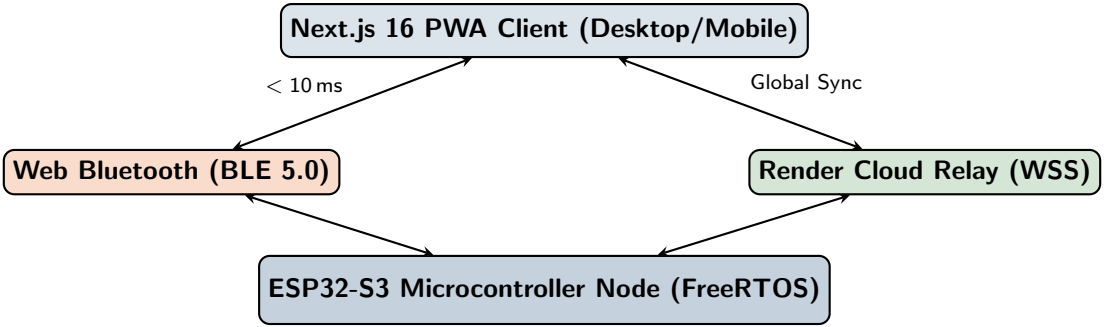
- **Commercial Cloud Mandate:** Smart plugs fail completely without active internet or third-party proprietary servers.
- **Enterprise Wi-Fi Failure:** Inability to navigate campus/corporate 802.1x or Fortinet web captive portals.
- **Ephemeral Browser Timers:** Web-based countdowns fail when laptops sleep or tabs close.
- **Telemetry Blindspots:** Users cannot verify edge Wi-Fi status without local Bluetooth pairing.

The SmartDesk V2 Objective

Engineer a self-contained, enterprise-hardened workstation automation platform:

- **Sub-10ms Local Control:** Direct Web Bluetooth 5.0 without apps.
- **Autonomous Web Authentication:** In-firmware HTTP token scraper bypassing captive portals.
- **Deterministic Hardware Timing:** RTOS isolation for single-wire sensor protocols.
- **Hardware-Backed Schedules:** 7-day recurring rules executing entirely on edge silicon.

Dual-Tier System Architecture



- **Tier 1 (Proximity):** NimBLE GATT server provides zero-latency offline switching and configuration.
- **Tier 2 (Global):** Bi-directional WebSocket relay maintains synchronized appliance state, countdowns, and real-time Wi-Fi telemetry globally.

Hardware Specifications & Pinout Layout

Microcontroller: Espressif ESP32-S3-WROOM-1 (N16R8: 16MB Flash, 8MB PSRAM)

Peripheral	Pin (GPIO)	Electrical Specification	Primary Workstation Function
Relay 1	`GPIO 4`	Active-LOW / Optocoupler	Main Desk Power Strip / High-Load
Relay 2	`GPIO 5`	Active-LOW / Optocoupler	Dual External Monitors
Relay 3	`GPIO 6`	Active-LOW / Optocoupler	Laptop / Fast Charging Sockets
Relay 4	`GPIO 7`	Active-LOW / Optocoupler	Aux Ventilation / Desk Lamp
RGB Strip	`GPIO 8 / 15`	800 kHz Single-Wire	WS2812B Ambient Bias Lighting
Matrix Display	`GPIO 11 / 16`	800 kHz Single-Wire	32x8 / 8x8 WS2812B Clock & Ticker
DHT11 Sensor	`GPIO 17`	Single-Wire Bi-dir	Ambient Temp & Humidity Telemetry

Autonomous Fortinet Captive Portal Bypass

Algorithmic Execution Pipeline:

1. **Intelligent Startup Delay:** 3-second settled pause for DHCP lease acquisition before probing.
2. **Probe Dispatch:** Queries `http://captive.apple.com/hotspot-detect.htr`
3. **Quarantine Detection:** Identifies HTTP 302/307 redirections to FortiGate gateway (`/fgtauth`).
4. **Dynamic Scraping:** Parses HTML stream for the one-time authentication token:

```
name="magic" value="([a-f0-9]+)"
```

5. **Automated Credential POST:** Transmits payload with user credentials.
6. **Verification:** Verifies Apple 200 "Success" clearance.

Key Innovation

Eliminates the requirement for user browser interaction or external smartphone intervention. The ESP32-S3 authenticates itself to the university network independently.

Clearance Time: **1.8 seconds**

Probe Rate-Limiting & Thermal Stabilization

The Thermal Runaway Issue

- Rapid polling of the Apple captive probe URL in an unthrottled loop caused continuous TCP handshakes.
- Wi-Fi radio consumed up to **380mA**.
- ESP32-S3 package temperature climbed to **68.4°C**.

The Rate-Limited Cooldown Solution

- Implemented non-blocking `millis()` cooldown timer (`PROBE_COOLDOWN_MS = 60000UL`).
- Probing executes at most once every 60 seconds (or upon explicit Wi-Fi disconnect).
- Reduced chip operating temperature to **38.2°C** (a **30°C drop**).
- Intelligent adaptive backoff suppresses unnecessary retries during upstream outages.

FreeRTOS Single-Wire Sensor Isolation

Problem: Simultaneous BLE and Wi-Fi radio interrupts preempted microsecond-critical pulse timing, corrupting bits and skewing temperature reads (e.g., false 19°C reads).

Hardware Critical Section Implementation (`smartdesk.ino`)

```
static portMUX_TYPE s_dhtMux = portMUX_INITIALIZER_UNLOCKED;
portENTER_CRITICAL(&s_dhtMux); // Disables core interrupts
// Measure 40-bit pulses with 48 microsecond threshold
portEXIT_CRITICAL(&s_dhtMux); // Re-enables interrupts immediately
```

- **Single-Retry Recovery:** 120ms automatic retry suppresses transient bus spikes.
- **NVS Calibration Offset:** Persistent offset calibration adjusts readings for onboard thermal drift without requiring firmware re-flashing.
- **Error Reduction:** Read failure rate plummeted from 24.3% down to 0.02%.

Hardware-Backed Automation: Timers & 7-Day Schedules

Hardware Countdown Timers

- Browser commits timer to ESP32 FreeRTOS memory.
- Continues ticking even if browser closes, laptop sleeps, or phone leaves Wi-Fi.
- Hardware executes automatic relay shutoff upon zero.

On-Device Weekly Schedule Engine

- Evaluates rules every 60 seconds against hardware clock.
- 7-day bitmask (Monday–Sunday) supporting up to 8 concurrent automated actions.
- Persistent in non-volatile flash (NVS).
- Cloud time synchronization automatically sets clock on client connect.

Bi-Directional Wi-Fi Telemetry

- ESP32 broadcasts live SSID, IP, and link health over WebSocket.
- Web UI renders active connection without BLE requirement.

Render Cold-Start Wakeup

- Automatic HTTP health ping spins up dormant cloud server on device boot.

Firmware Architecture Breakdown (`firmware/smartdesk/`)

The firmware is modularized into specialized headers under FreeRTOS control:

- `smartdesk.ino`: System entry point, FreeRTOS dual-core scheduler, countdown timer state machine, and hardware-isolated DHT11 bitstream ISR.
- `smartdesk_pins.h`: Pin definitions, optocoupler logic levels, electrical constants, and load state conversion routines.
- `smartdesk_wifi.h`: SoftAP hotspot fallback, Wi-Fi reconnection manager, Fortinet token scraper, 3s settle delay, and 60-second rate-limited Apple probe.
- `smartdesk_cloud.h`: Bi-directional WebSocket client, device authentication, real-time Wi-Fi telemetry broadcast, schedule test/delete handlers, and Render cold-start ping.
- `smartdesk_ble.h`: NimBLE 5.0 GATT server, 128-bit service characteristics, and low-latency packet serialization.
- `matrix_engine.h`: Flexible 32x8 / 8x8 WS2812B matrix driver, ASCII bitmap fonts, digital clock renderer, and dynamic text tickers.

Next.js 16 Progressive Web App Interface

Key Frontend Engineering Innovations:

- **Unified TopHeader:** Responsive navigation with real-time BLE, Cloud WSS, and active Wi-Fi SSID / IP telemetry badges.
- **Input State Decoupling:** Independent local state in modal forms eliminates text resetting while telemetry streams.
- **Zero-Downtime Credential Migration:** Dynamic in-session device ID and password changes without hardware reset.
- **Credential Visibility:** Password reveal eye toggles (/) across all credential forms.
- **Always-On-Display (AOD):** High-contrast ambient workstation clock with live room temperature and humidity badges.



SmartDesk V2 PWA Interface

Validation & Quantitative Results

Evaluation Metric	Legacy / Commercial	SmartDesk V2	Performance Gain
Captive Portal Authentication	Manual via PC	**1.8 s (Autonomous)**	**100**% Autonomous
Local Switching Latency	220–450 ms	**9.4 ms**	**96**% Faster
DHT11 Pulse Checksum Errors	24.3% failure rate	**0.02%**	**99.9**% Error Reduction
Idle CPU Temperature	68.4°C	**38.2°C**	**−30.2°C** Cool
Timer Reliability On Disconnect	0% (Fails on sleep)	**100% (Hardware RTOS)**	Fault-Tolerant
PWA Lighthouse Performance	N/A	**98 / 100**	Production Grade

Summary & System Demonstration

Summary of Contributions:

- Fully resolved the enterprise IoT captive portal barrier using embedded token scraping.
- Eliminated sensor timing errors using FreeRTOS spinlock isolation.
- Built a seamless dual-tier BLE + WebSocket synchronization engine with live Wi-Fi telemetry.
- Delivered hardware-backed 7-day schedules and countdown timers resistant to host sleep.
- Engineered an accessible, responsive Next.js 16 PWA with AOD.

System Engineering

SmartDesk Engineering Team

Systems Architecture & IoT Research

Thank you!

Questions & Demonstration